

CHAPTER 7: **PRIVACY**

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OBJECTIVES

- Define privacy
- Know threat to privacy
- Know how to protect privacy



INTRODUCTION

- Information collection, exchange, combination, and distribution easier than ever
- More information access → less privacy
- Trade-offs

Privacy vs need for credentials

Privacy vs desire for free expression

Privacy vs safety/security

DEFINITION

- The right of privacy
 - the right to be left alone
- The right of individuals to control the collection and use of information about themselves



DEFINITION (CONT..)

■ Key aspects

- 1) freedom from intrusion – being left **alone**
- 2) **control** of information about oneself
- 3) **freedom** from surveillance (from being followed, watched, and eavesdropped)

HARMS OF PRIVACY

- Too much individual privacy can harm society
- Cover for illegal or immoral activities
- Burden on the nuclear family
- Hidden dysfunctional families
- Ignored people on society's fringes

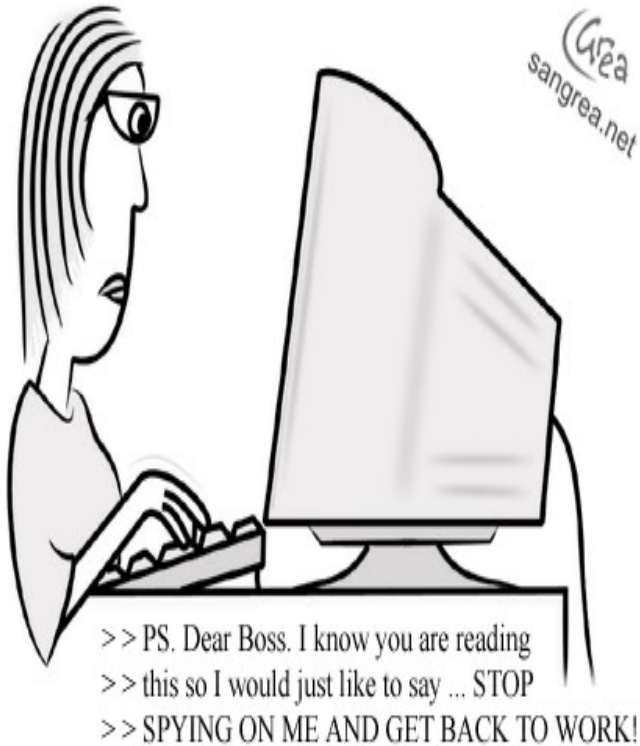
BENEFITS OF PRIVACY

- Individual growth
- Individual responsibility
- Freedom to be yourself
- Intellectual and spiritual growth
- Development of loving, trusting, caring, intimate relationship

RISKS OF THE TECHNOLOGY

- **Invisible information gathering**
 - collection of personal information about someone without the person's knowledge.
- **Ethical issue is :**
 - if someone is not aware that the information is being collected or how it will be used, he or she has no opportunity to consent or withhold consent for its collection and use.

RISKS OF THE TECHNOLOGY (CONT..)



- Satellite surveillance
- Automatic identification of a person's telephone number for toll free call
- Supermarket club cards
- ISP knows every site we visit
- Cookies

RISKS OF THE TECHNOLOGY (CONT..)

- **Secondary use of personal information**
 - use of information for a purpose other than the one for which it was supplied.
 - difficult for us to control our personal information if it is collected by business, organization and share with or sold to others
 - eg. Sale of consumer information to marketers or other business, use of information to deny job, deny bank loan (eg: CTOS –Credit Tip Off Service Sdn. Bhd)

RISKS OF THE TECHNOLOGY (CONT..)

- **Computer matching**
 - combining and comparing information from different databases
 - eg businesses use the technique to form consumer dossiers, government uses to detect fraud
- **Computer profiling**
 - using data in computer files to determine characteristics of people most likely to engage in certain behavior
 - eg determination of customers for specific product

RISKS OF THE TECHNOLOGY (CONT..)

■ Location

- many devices allow our location to be determined
- eg. GPS, satellite, computer chip, wireless watchband transmitter



CASE STUDY: SECRET MONITORING

- **Sullivans have a baby girl**
- **Both work; they are concerned about performance of full-time nanny**
- **Purchase program that allows monitoring through laptop's camera placed in family room**
- **They do not inform nanny she is being monitored**

RULE UTILITARIAN EVALUATION

- If everyone monitored nannies, it would not remain a secret for long
- Consequences
 - Nannies would be on best behavior in front of camera
 - Might reduce child abuse and parents' peace of mind
 - Would also increase stress and reduce job satisfaction of child care providers
 - Might result in higher turnover rate and less experienced pool of nannies, who would provide lower-quality care
- Harms appear greater than benefits, so we conclude action was wrong

SOCIAL CONTRACT THEORY EVALUATION

- It is reasonable for society to give people privacy in their own homes
- Nanny has a reasonable expectation that her interactions with baby inside home are private
- Sullivan's decision to secretly monitor the nanny is wrong because it violates her privacy

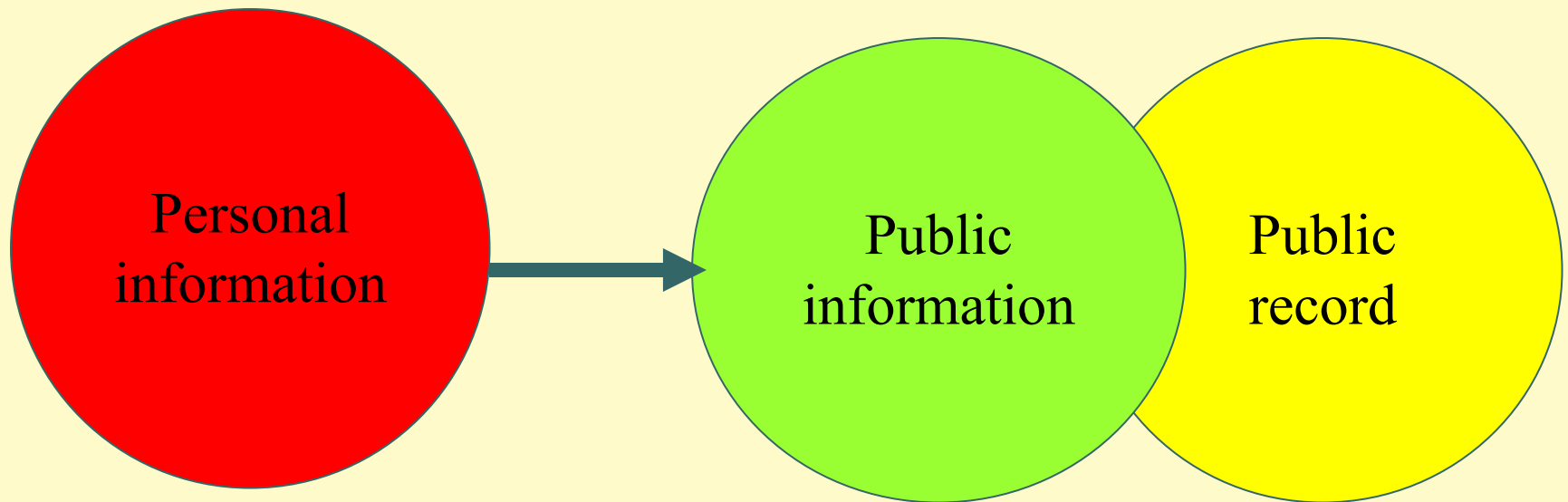
KANTIAN EVALUATION

- Imagine rule, “An employer may secretly monitor the work of an employee who works with vulnerable people”
- If universalized, there would be no expectation of privacy by employees, so secret monitoring would be impossible
- Proposed rule is self-defeating, so it is wrong for Sullivans to act according to the rule

SUMMARY

- Three analyses have concluded Sullivans were wrong to secretly monitor how well their nanny takes care of their baby
- Morally acceptable options
 - Conduct more comprehensive interview of nanny
 - More thoroughly check nanny's references
 - Spend a day or two at home observing nanny from a distance
 - Be up-front with nanny about desire to install and use surveillance software on laptop

DISCLOSING INFORMATION



DISCLOSING INFORMATION

- **Public record** : information for public access.
Eg birth certificate, marriage licenses, motor vehicle records, criminal record
- **Public information** : information revealed to an organization that has right to share it. Eg telephone directory
- **Personal information** : undisclosed information

DATA GATHERING AND PRIVACY IMPLICATIONS

- Facebook tags
- Enhanced 911 services
- Rewards or loyalty programs
- Body scanners
- RFID tags
- Implanted chips
- Cookies and flash cookies
- Etc.



FACEBOOK TAGS

- Tag: Label identifying a person in a photo
- Facebook allows users to tag people who are on their list of friends
- About 100 million tags added per day in Facebook
- Facebook uses facial recognition to suggest name of friend appearing in photo
- Does this feature increase risk of improper tagging?

ENHANCED 911 SERVICES

- Cell phone providers in United States required to track locations of active cell phones to within 100 meters
- Allows emergency response teams to reach people in distress
- What if this information is sold or shared?

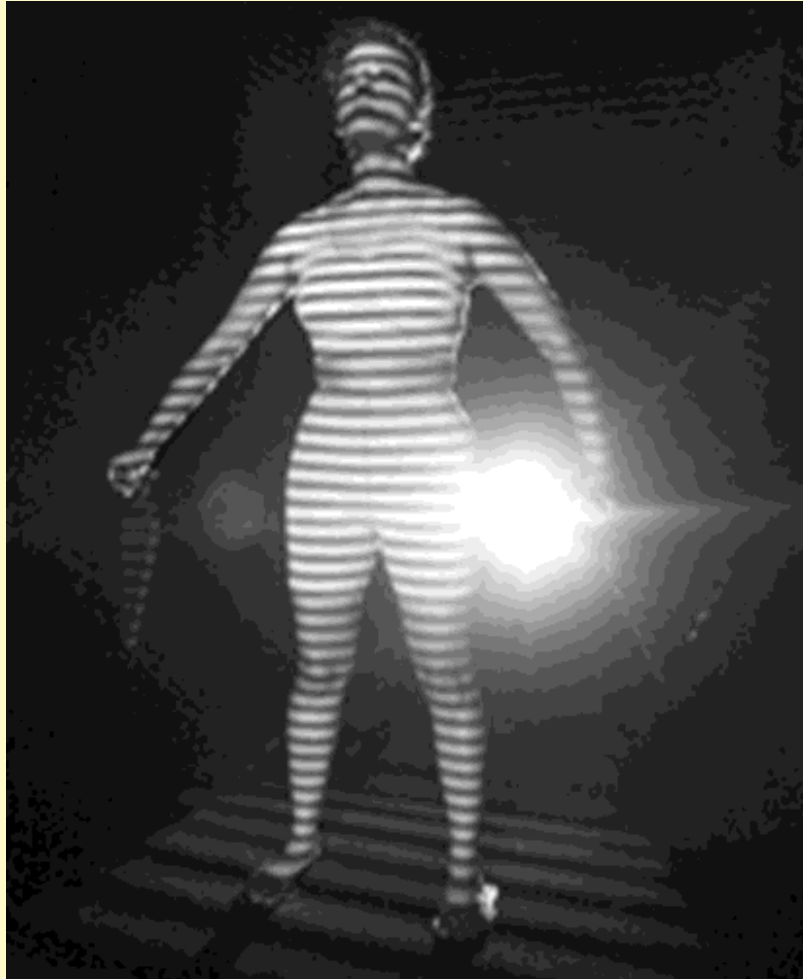
REWARDS OR LOYALTY PROGRAMS

- Shoppers who belong to store's rewards program can save money on many of their purchases
- Computers use information about buying habits to provide personalized service
 - ShopRite computerized shopping carts with pop-up ads
- Do card users pay less, or do non-users get overcharged?

BODY SCANNERS

- Some department stores have 3-D body scanners
- Computer can use this information to recommend clothes
- Scans can also be used to produce custom-made clothing

BODY SCANNER TAKES MEASUREMENTS



RFID TAGS

- **RFID: Radio frequency identification**
- **An RFID tag is a tiny wireless transmitter**
- **Manufacturers are replacing bar codes with RFID tags**
 - **Contain more information**
 - **Can be scanned more easily**
- **If tag cannot be removed or disabled, it becomes a tracking device**

IMPLANTED CHIPS

- Taiwan: Every domesticated dog must have an implanted microchip
 - Size of a grain of rice; implanted into ear
 - Chip contains name, address of owner
 - Allows lost dogs to be returned to owners
- RFID tags approved for use in humans
 - Can be used to store medical information
 - Can be used as a “debit card”

DIGITAL VIDEO RECORDERS

- TiVo service allows subscribers to record programs and watch them later
- TiVo collects detailed information about viewing habits of its subscribers
- Data collected second by second, making it valuable to advertisers and others interested in knowing viewing habits

COOKIES

- **Cookie:** File placed on computer's hard drive by a Web server
- **Contains information about visits to a Web site**
- **Allows Web sites to provide personalized services**
- **Put on hard drive without user's permission**
- **You can set Web browser to alert you to new cookies or to block cookies entirely**

FLASH COOKIES

- Flash cookie: File placed on your computer's hard drive by a Web server running the Adobe Flash Player
- Flash cookie can hold 25 times as much information as a browser cookie
- Flash cookies not controlled by browser's privacy controls
- Some Web sites use flash cookies as a way of backing up browser cookies. If you delete browser cookie, it can be "respawned" from the flash cookie
- Half of 100 most popular Web sites use flash cookies

DISCLOSING INFORMATION

- Types of disclosures:
 - Voluntary
 - Involuntary
 - Statutory

PRINCIPLES FOR DATA COLLECTION AND USE

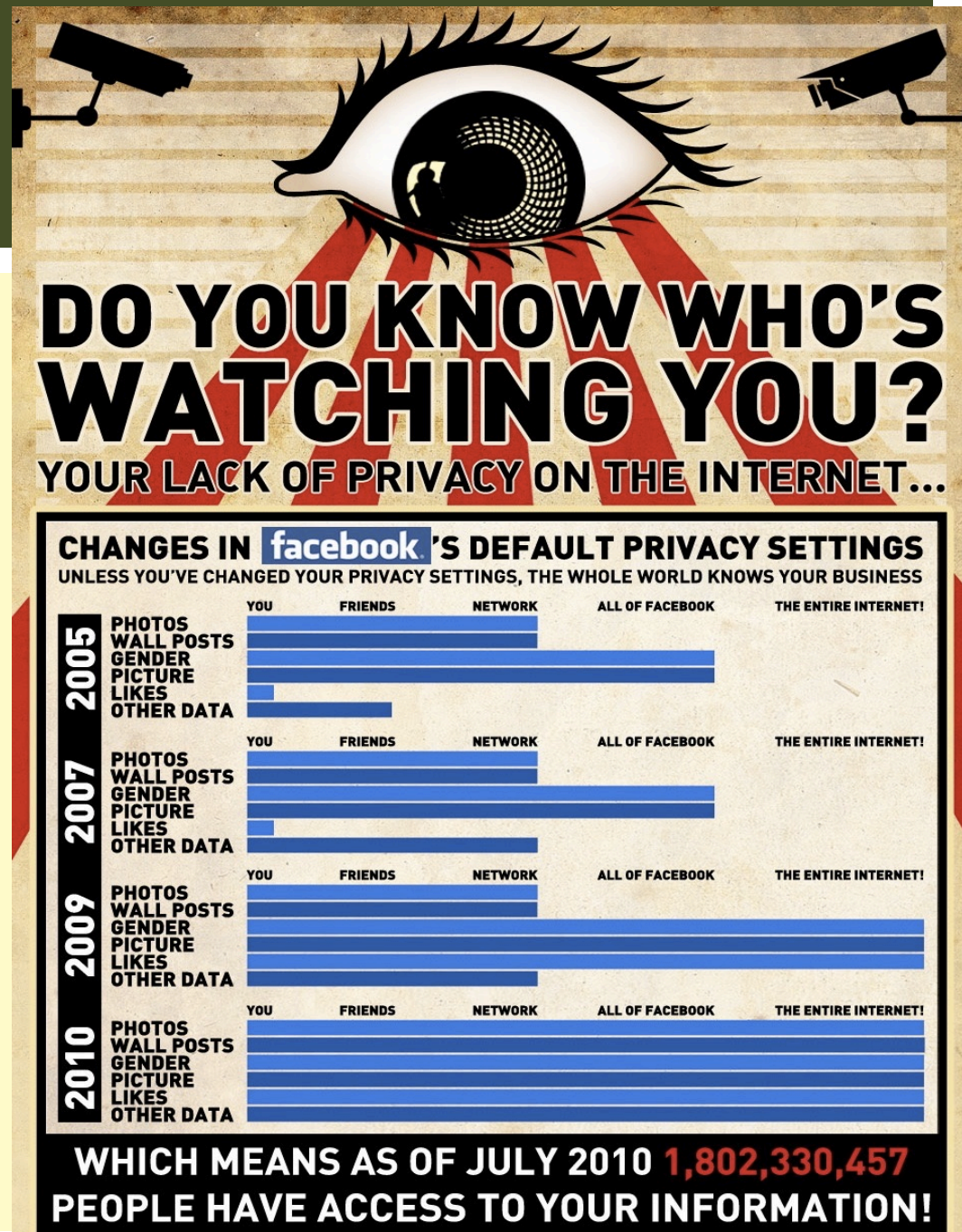
- Informed consent
 - business and organization must inform consumers about what information they are collecting and how they will use it
- Offer a way for people to opt out from mailing lists and from transfer of their data to other parties
- Provide stronger protection for sensitive data.
Eg: use an opt in policy for disclosure of medical data

PRINCIPLES FOR DATA COLLECTION AND USE

- Collect only the data needed
- Keep data only as long as needed
- Maintain accuracy and security of data
- Provide a way for people to access and correct data stored about them
- Sanction against employees who release information without authorization

PRIVACY PROTECTION

- Awareness
 - how technology works
 - how it is being used
 - what the risks are
 - what tools are available to reduce exposure and unwanted uses of personal data



PRIVACY PROTECTION

- Awareness as consumers
 - decide to what extent to use privacy protection tools
 - be more careful about information given out
 - consider the privacy policies of business and web sites we use or visit

PRIVACY PROTECTION

- Awareness as business managers
 - learn and implement techniques to respond to the privacy demands of consumers
- Awareness as computer professional
 - design database systems/web software that make it easier to build in privacy protection and reduce the risks of unauthorized leaks

PRIVACY PROTECTION

■ Privacy-Enhancing Technologies

- cookie disablers
- alerting users whenever a web site is about to store a cookie and allow them to reject it
- surfing web anonymously, leaving no record of visiting site; Anonymizer.com

PRIVACY PROTECTION

- Privacy-Enhancing Technologies
 - Platform for Privacy Protection (P3P)
‘automatically’ protect privacy by user specifying his/her privacy requirement
 - a well-designed database to protect leak, intruders, unauthorized access
 - => restriction of operation
 - => audit trail to trace unauthorized activities
 - => encrypted information

PRIVACY PROTECTION

- **Trusted Third Parties**
 - firms whose reputation for honesty is important to their business to process confidential data
- **Paying for Consumer Information**
 - willing to trade some privacy for other things
- **Explicit policy statement**
- **Opt in opt out policy**

PRIVACY PROTECTION

■ Market Response

market can protect privacy by:

=> using trusted third parties

=> adhering to established privacy policy

=> purchasing consumer information directly from the consumers

=> developing and selling privacy-enhancing technology and services

PRIVACY PROTECTION: REGULATION

- Basic legal framework
 - => enforcement of agreements and contracts
 - freedom to create them and enforcement of their terms by the legal system
 - => publish privacy policy



PRIVACY PROTECTION: REGULATION

- Basic legal framework
 - => set privacy-protection default not in contracts
 - information could be used only for the direct and obvious purpose for which it was supplied

PRIVACY PROTECTION: REGULATION

- Requiring specific consent policies
 - => adhere to informed consumer consent
 - must clearly state policy for use of personal information. If a person proceeds and purchase, explores the web sites, provides information, and so on, consent to the policy is assumed
 - must provide an opt out policy
 - must use an opt in policy
 - must obtain consent for each individual secondary use, disclosure, or transfer of their personal information

MAYBANK2U

■ Privacy Policy

- Our customers have always been our top priority and the Maybank Group is dedicated to ensuring that our online financial transactions are private, safe and secure. In short, your privacy is our top priority.
- This privacy policy is designed to explain some of the privacy principles we abide by, in providing a responsible and reliable online experience



MAYBANK2U

■ Sharing Information

- The only other instances where we disclose your information is:
 - **When it is required by law**
When such disclosure is required under legislation or regulation, government agency or court order.
 - **When it involves public interest**
When such disclosure is required to protect the interest of the public, for example in the detection of crimes, etc.
 - **Where it is authorized by you**

DISCUSSION QUESTION

- Music files downloaded from Apple's iTunes Store have the purchaser's name and email address embedded in them. Conceivably, Apple could use this information to learn how much file sharing goes on (e.g., it could find out that a month after Ann purchases a song there are 10 computers that have a copy of Ann's music file)
- By including personal information in music files it sells, has Apple violated the privacy rights of its customers?